Project Brief: Fernhill Software, version 2 of 3 (the trial designer's model)

1. The organization

Fernhill Software is a useful fiction, a business-software firm with about 1,100 engineers in 120 product teams. It hires engineers on technical skill and promotes its strongest into management, where some struggle. In 2025, as AI coding tools arrive, an AI enthusiast on the leadership team proposes a three-part reorganization. First, hire professional managers on people skill instead of promoting engineers. Second, make each team's engineer-manager its technical lead, keeping technical judgment but managing no one. Third, let AI fill the manager's technical gaps. **Fernhill runs a one-year pilot** from July 2025, assigning its 120 teams by lot to three arms of 40 so the reorganization and AI can be told apart.

2. The model

A team delivers when two jobs get done, coordinating people and resolving technical unknowns. The proposal splits those jobs between a manager and a lead, with AI assisting both.

1. **People skill lowers coordination friction.** Less work sits blocked on other people, more so in bigger teams.
2. **Difficulty sets how many unknowns arrive.** AI resolves routine ones but helps little on hard ones.
3. **Escalated unknowns queue at whoever holds technical judgment**, the engineer-manager or a tech lead with far more time. Near capacity, the wait explodes (Little's Law).

In short, AI should help most on easy work, and the new structure most on hard work, where queues form.

For the model to be right, the professional managers must have more people skill than the engineer-managers they replace. Teams must stay in their arm, and teams without AI must go without it. The leads must accept their new role. Difficulty must be judgeable at quarterly planning. **What stays unmeasured** is the judgment holder's capacity, which shows only through waits.

3. The question

Should Fernhill replace engineer-managers with professional managers backed by technical leads and AI? The pilot answers with two comparisons, with each arm also set against its own eight pre-pilot quarters. The first compares **AI, holding structure fixed** (engineer-managers with and without AI). The second compares **structure, holding AI fixed** (manager + lead vs. engineer-manager, both with AI), the decision Fernhill faces.

4. The prediction

Stated before looking at this version's data:

- **AI alone** raises delivery on easy quarters by about 8 points, and cuts their escalations by about 40%, but changes little on hard quarters.
- **People-skill ratings** are higher under professional managers.
- **The new structure** shortens escalation waits, most on hard quarters, and on hard quarters delivers about 5 points more than engineer-managers with AI.

The model is wrong if any of these holds:

- AI's gain is as large on hard quarters as on easy ones.

- The new structure's waits are no shorter on hard quarters.
- Its waits are shorter but its delivery is no better. Then the queue isn't what limits delivery.

5. The data sketch

The data. Fernhill's records have one row per team per quarter, July 2023 to June 2026 (1,440 rows), and they are simulated, since Fernhill is fictional. **The variables,** by concept. The change is arm (assigned by lot), manager type, and AI licenses, with IT's usage logs. The manager has a people-skill rating on the eight Project Oxygen behaviors, and the hours work sits blocked on other teams. The work has team size, and difficulty rated at planning (a rating new with the pilot). Unknowns are escalations, and the hours each waits. The outcome is the share of committed milestones delivered on time.

6. The disciplinary anchor

Whom to promote (organizational management). Firms promote their best performers, who then make worse managers (Peter and Hull, 1969; Benson, Li and Shue, 2019). People skill keeps teams together, and Google's Project Oxygen ranked technical skill last of eight manager behaviors (Hoffman and Tadelis, 2021; Garvin, 2013).

The work itself (operations management). Unknowns go up the hierarchy unless knowledge is cheap, and AI makes only routine knowledge cheap (Galbraith, 1974; Garicano, 2000; Brynjolfsson, Li and Raymond, 2025; METR, 2025). Waits grow when escalations arrive faster than they are cleared (Little's Law).

The method. The pilot is randomized, so the lot makes the arms comparable and their differences are the effects of AI and of the structure.

Note on AI use

I used Gemini (Willamette account) to turn the critique of version 1 into a pilot design and size its arms. It suggested a fourth arm, a professional manager without AI. I rejected it, since three arms answer the decision Fernhill faces. It suggested surveying how managers and leads work together. I left that out, since the escalation wait already measures how fast technical questions get answered.